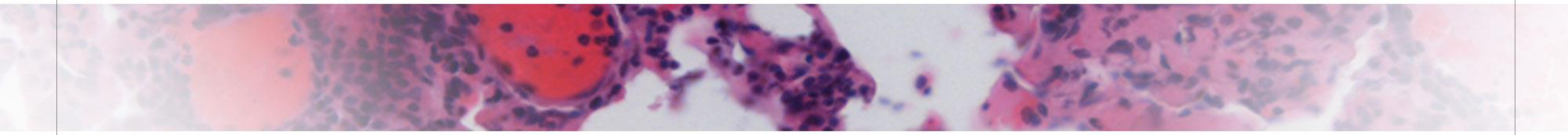
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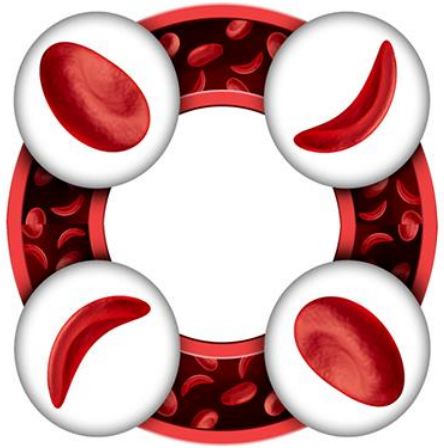


Cerebral and peripheral microvascular function in sickle cell disease: Blood flow and oxygen extraction via diffuse correlation and Near-Infrared spectroscopy

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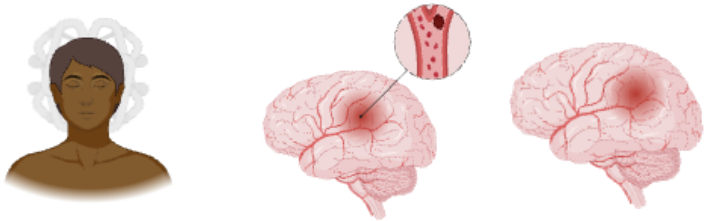
Sickle Cell Disease



- Sickle cell disease (SCD), an inherited blood disorder, impedes microvascular blood flow
  - Affects 20 million individuals globally

- Patients with Sickle Cell Disease (SCD) experience:
  - Chronic anemia and vaso-occlusion
  - leading to reduced oxygen delivery

Complications of SCD

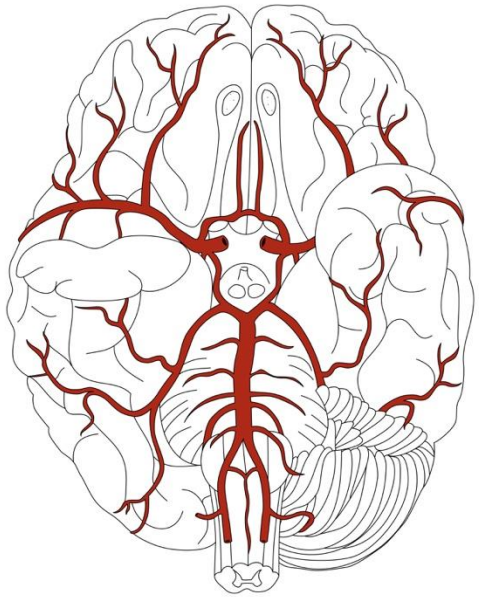


Cognitive Impairment | **Silent Cerebral Infarction** | **Stroke**

Cerebral Hemodynamics

- **Oxygen Extraction Fraction (OEF):**
  - Fraction of oxygen absorbed by the tissue from blood
  - Elevated in pediatric SCD populations

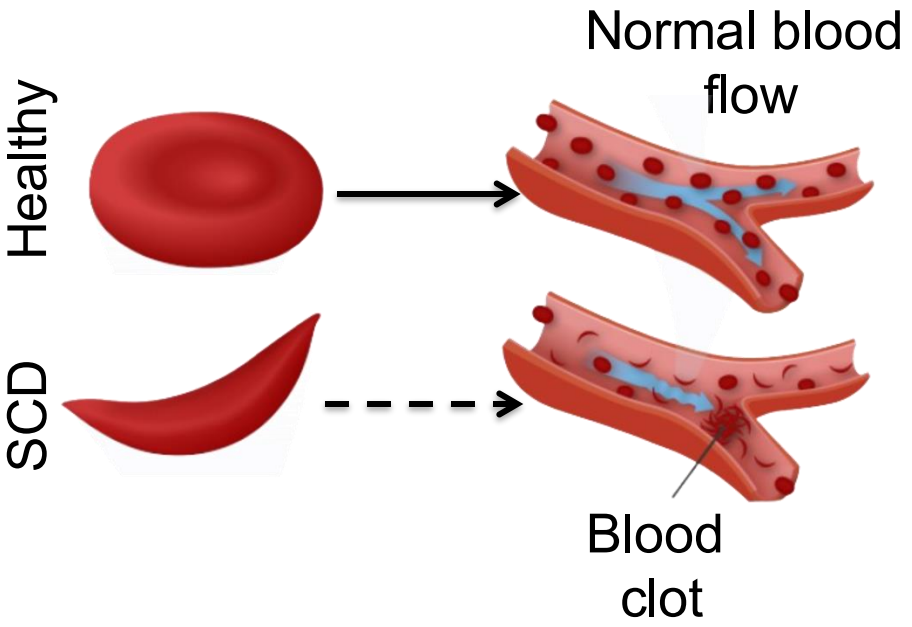
- **Cerebral tissue oxygen saturation (StO2):**
  - Percentage of dissolved oxygen in cerebral tissue
  - A critical endpoint for monitoring healthy tissue perfusion in SCD



Knowledge Gap

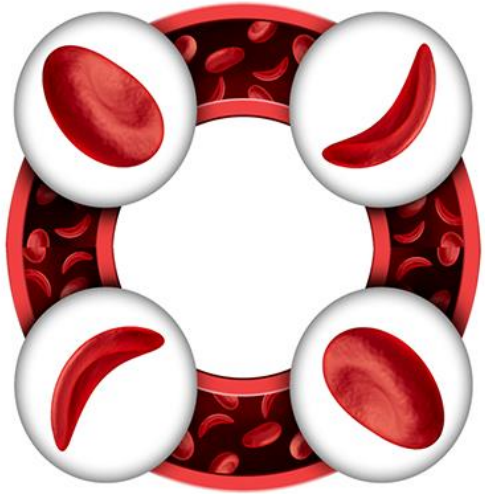
Despite evidence of altered cerebral oxygen extraction in SCD, the relationship between **cardiac pulsatility** and **regional oxygen supply demand** has not been quantified.

Understanding this interaction could reveal how **vascular regulation differs between cerebral and peripheral tissues** in adults with SCD.



Novelty

The first study to quantify cardiac pulsatility metrics (Peak Systolic Flow(PSF), End Diastolic Flow(EDF), Pulsatility Index(PI), Area Under the Curve(AUC)) alongside cerebral and peripheral oxygen extraction in adults with SCD, providing insight into how pulsatile blood flow influences regional oxygen supply–demand balance.

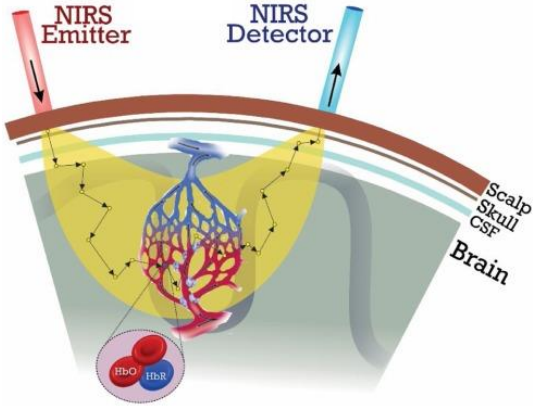


Goal of the Study

To investigate how **cardiac pulsatility (PSF, EDF, PI, AUC)** is influenced by regional oxygen supply-demand in adults with SCD compared to controls, using **diffuse correlation (DCS)** and **near-infrared spectroscopy (NIRS).**

fNIRS and DCS

- **Functional Near-Infrared Spectroscopy (fNIRS):**
  - Assesses changes in tissue oxygenation using NIR light
  - Measures **oxyhemoglobin (HbO)**, **deoxyhemoglobin (Hb)**, and **total hemoglobin (HbT)**
  - Provides **tissue oxygen saturation (StO_2)**, reflecting oxygen delivery and utilization
- **Diffuse Correlation Spectroscopy (DCS):**
  - Measures microvascular **blood flow dynamics** non-invasively
  - Sensitive to **temporal fluctuations in scattered light intensity** caused by red blood cell motion

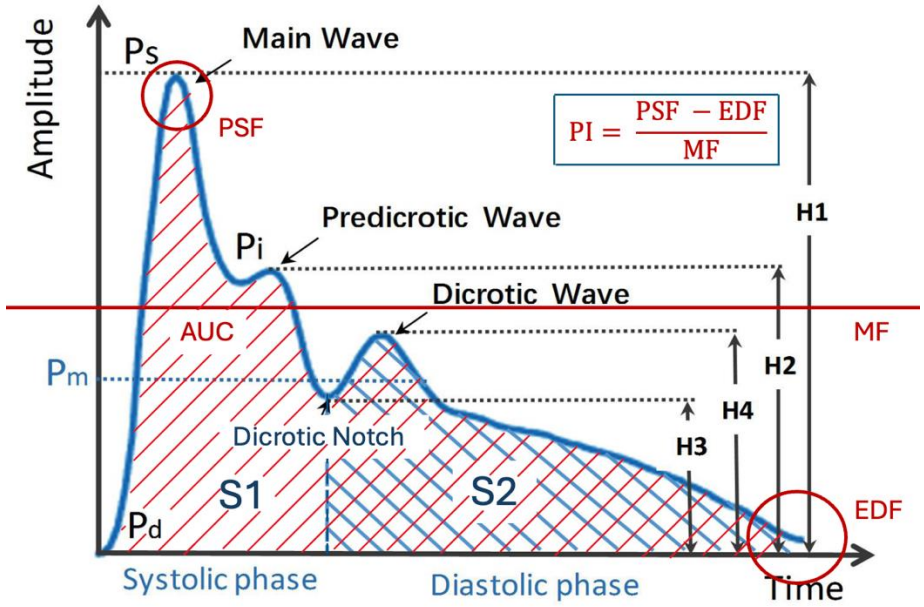


Mohammadi-Neja et al, 2018

Cardiac Pulse

- **Using DCS and NIRS, we assessed:**
  - ***Cerebral and peripheral blood flow dynamics:***
    PSF, EDF, PI, AUC
  - ***Oxygenation metrics:***
    OEF and StO_2

- **Cardiac Pulsatility Insight:**
  - Pulsatile waveform from DCS reflects fluctuations in scattered light caused by red blood cell motion, enabling estimation of blood flow dynamics.



Hypothesis

Adults with SCD would exhibit **altered blood flow pulsatility and elevated OEF** compared to controls, reflecting impaired microvascular function due to chronic anemia and vaso-occlusion.

Methods

Study Participants:

- 45 adults enrolled from UPMC
- **SCD (n = 21)**: HbSS/HbSC genotypes
- **Controls (n = 24)**: HbAA/HbAS genotypes

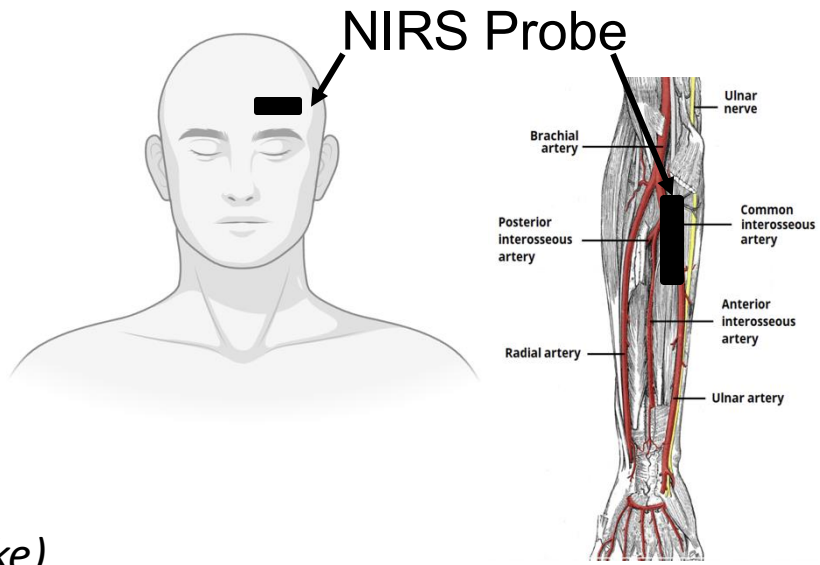
Measurement Sites:

Cerebral (Prefrontal) cortex, Forearm muscle

Exclusion criteria:

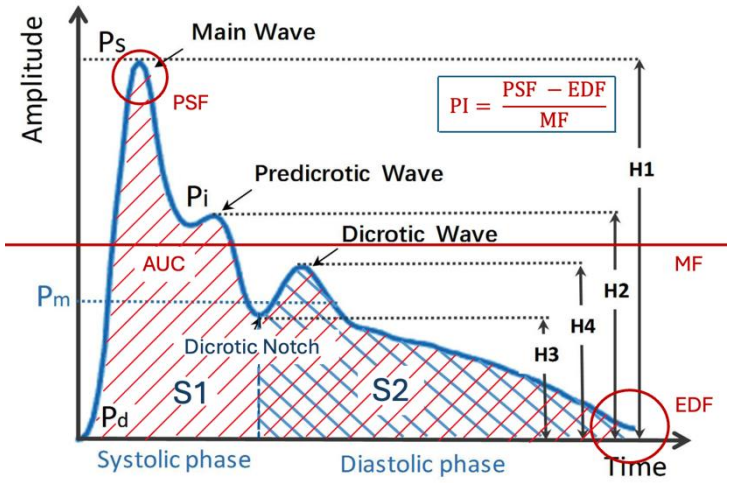
For controls only:

- *Disorders affecting blood vessels (e.g. stroke)*
- *Caffeine / nicotine / alcohol drugs for 3 hours*
- *NSAIDs / aspirin for 7 days prior to study visit*



Extracted Parameters from Cardiac Waveform

Peak Systolic Flow (PSF)	Maximum flow during a cardiac cycle
End Diastolic Flow (EDF)	Minimum flow during a cardiac cycle
Pulsatility Index (PI)	Relative change in flow amplitude
Area Under the Curve (AUC)	Blood volume per cardiac cycle

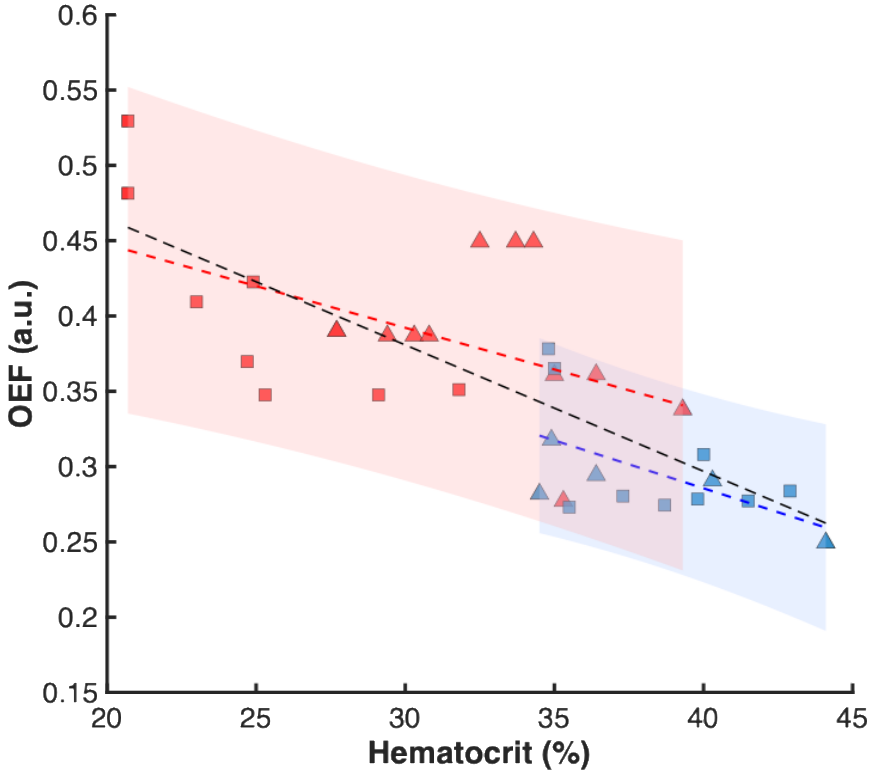


These parameters capture the dynamics of blood flow within the cardiac cycle and were analyzed alongside OEF, StO_2, and hematocrit (Hct) to understand **how pulsatile blood flow supports or challenges oxygen delivery** in adults with SCD.

Results

- Adults with SCD exhibit higher OEF and a strong inverse relationship between OEF and Hct, reflecting compensatory oxygen extraction in SCD anemia.

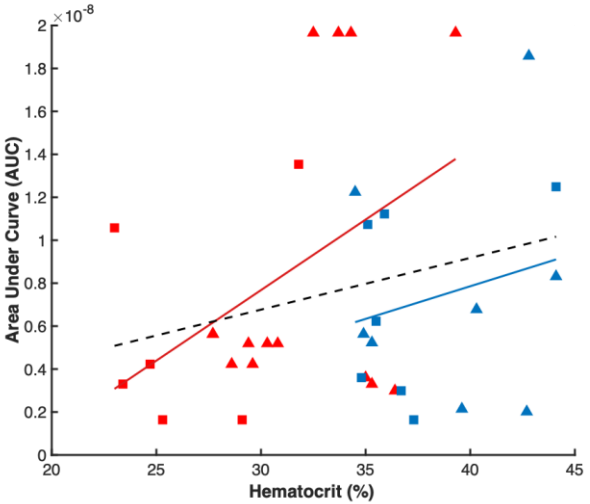
Parameter	Control (n=15)	SCD(n=20)
p-value	0.031	0.021
r-value	-0.58	-0.51



Results

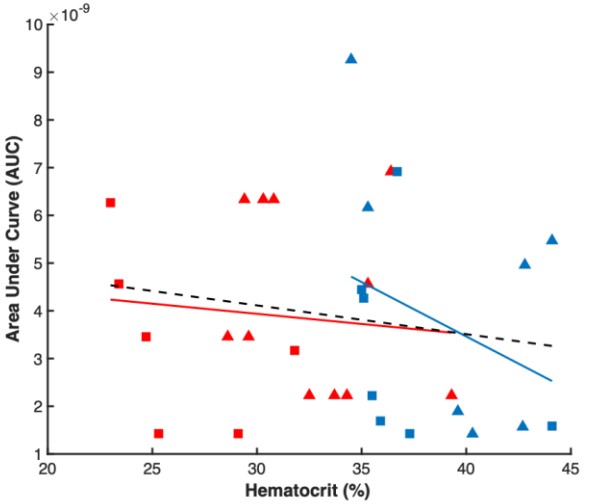
- Opposite trends between cerebral and peripheral cardiac blood volume regions suggest differential vascular control in brain compared to the arm when measured with AUC

CEREBRAL REGION



Parameter	Control (n=15)	SCD (n=20)
p-value	0.415	0.051
r-value	0.23	0.44

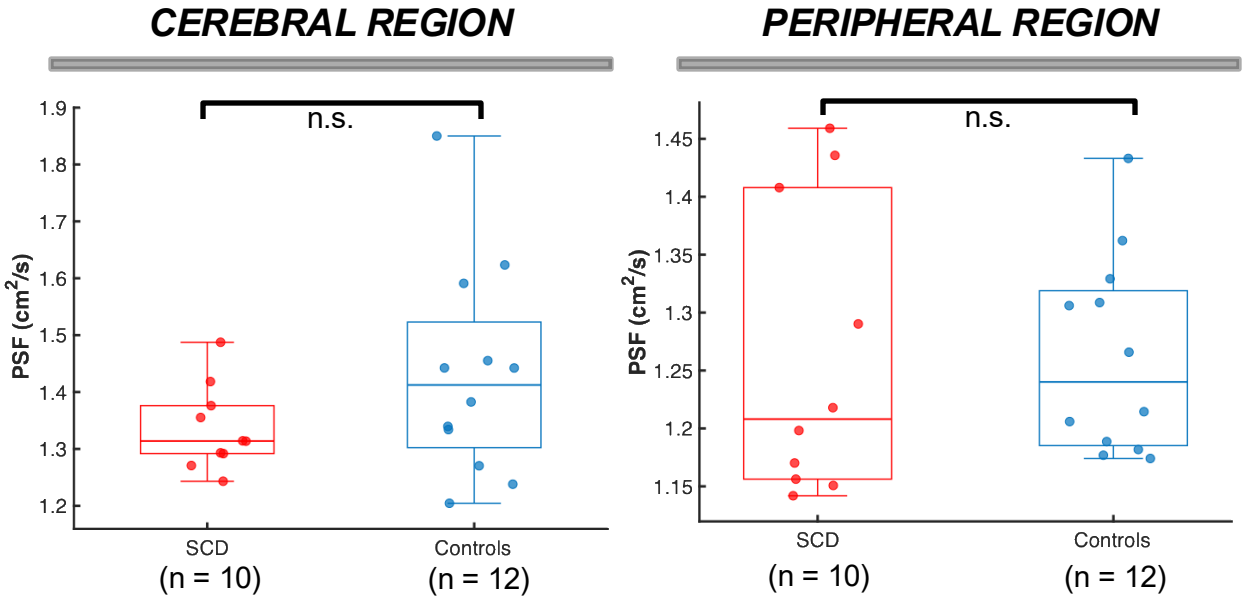
PERIPHERAL REGION



Parameter	Control (n=14)	SCD (n=17)
p-value	0.245	0.694
r-value	-0.33	-0.10

Results

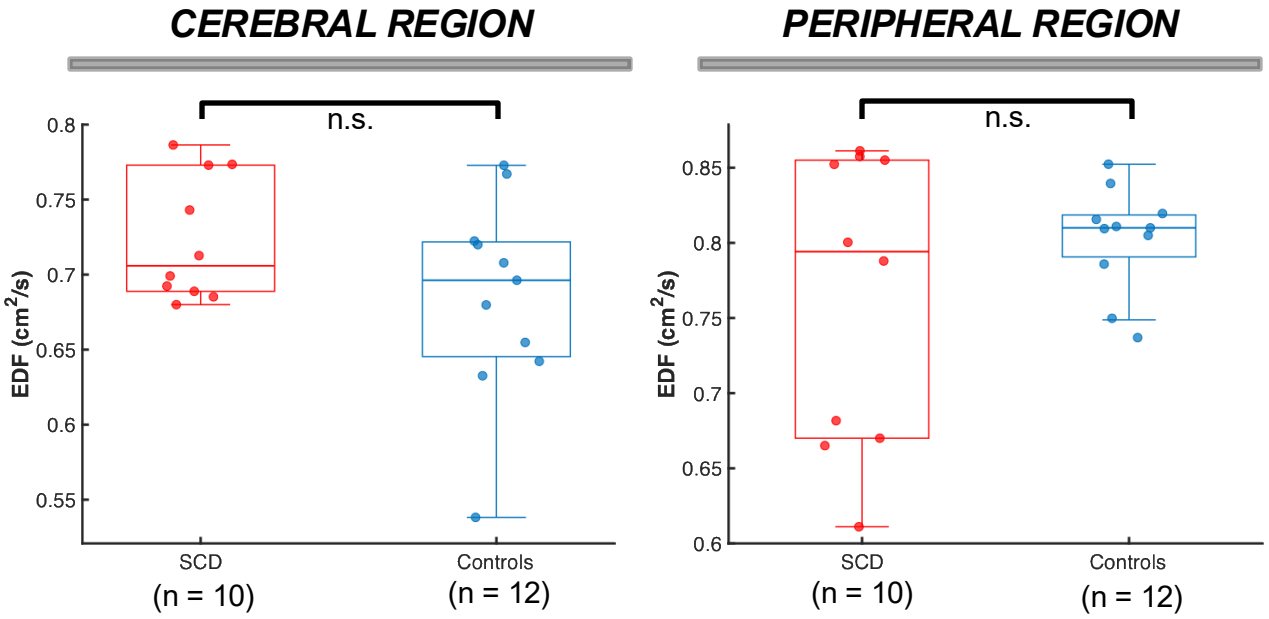
We found that those with SCD exhibited a wider observed range (PSF: IQR Range = 1.15 to 1.4) and greater variability of blood flow parameters compared to controls (PSF: IQR = 1.19 to 1.31).



Although not statistically significant, SCD participants showed greater variability in peripheral flow, indicating heterogeneous microvascular function, while cerebral measures showed narrower variability, suggesting preserved autoregulation despite anemia.

Results

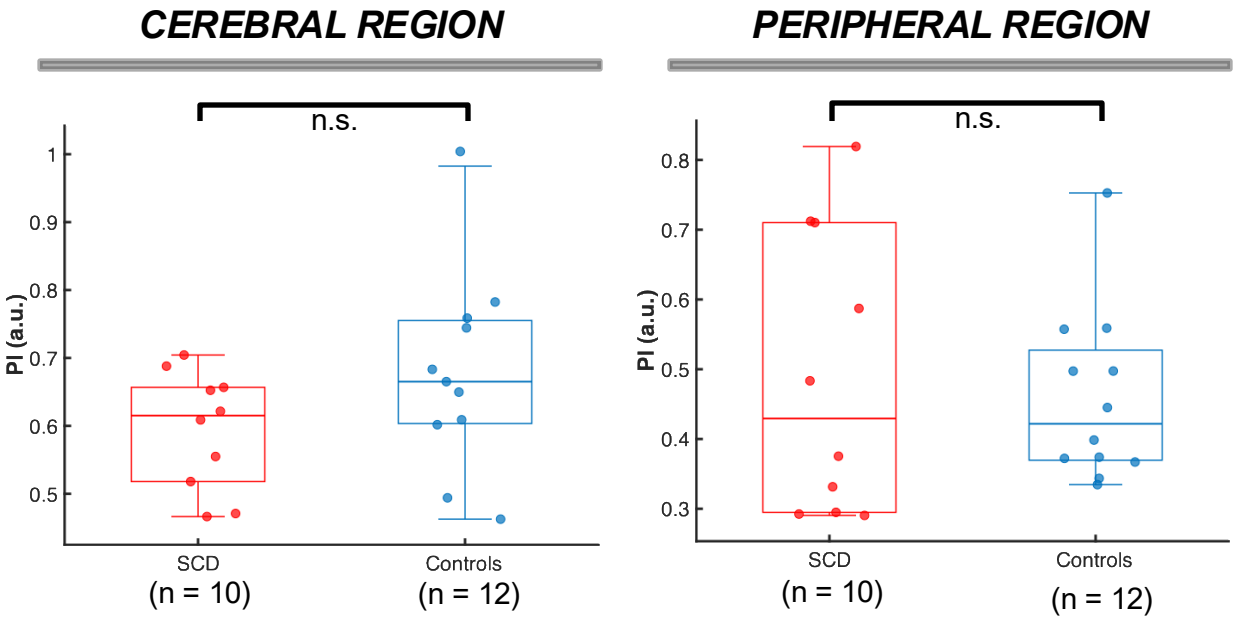
We found that those with SCD exhibited a wider observed range (EDF: IQR Range = 0.66 to 0.85) and greater variability of blood flow parameters compared to controls (EDF: IQR = 0.79 to 0.83).



Although not statistically significant, SCD participants showed greater variability in peripheral flow, indicating heterogeneous microvascular function, while cerebral measures showed narrower variability, suggesting preserved autoregulation despite anemia.

Results

We found that those with SCD exhibited a wider observed range (PI: IQR: 0.3 to 0.7) and greater variability of blood flow parameters compared to controls (PI: IQR : 0.38 to 0.52).

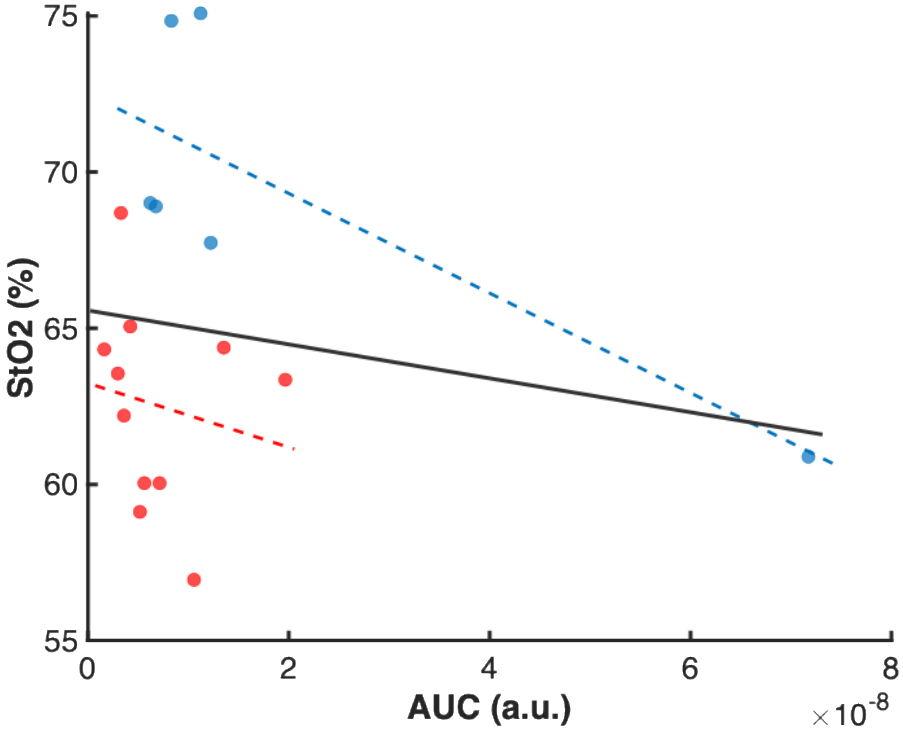


Although not statistically significant, SCD participants showed greater variability in peripheral flow, indicating heterogeneous microvascular function, while cerebral measures showed narrower variability, suggesting preserved autoregulation despite anemia.

Results

- AUC showed weak, non-significant negative correlations with StO_2 in both groups, which suggests reduced cerebral flow-oxygen coupling in SCD

Parameter	Control (n=6)	SCD (n=11)
p-value	0.065	0.616
r-value	-0.784	-0.171



Discussion and Conclusion

- OEF was significantly higher in SCD and showed a strong negative correlation with hematocrit, indicating that reduced oxygen-carrying capacity drives greater oxygen extraction to maintain metabolic demand.
- Weak, non-significant associations between AUC and Hct suggest differential vascular regulation across regions, with stable cerebral flow versus heterogeneous peripheral flow in response to anemia.
- Weak, non-significant inverse trends between AUC and StO_2 suggest inefficient coupling between pulsatile flow and oxygenation in adults with SCD.
- SCD showed wider variability in pulsatility metrics, reflecting heterogeneous microvascular function and impaired flow stability in periphery

Acknowledgements

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Poster Presentation

Session Name: 113. Sickle Cell Disease, Sickle Cell Trait, and Other Hemoglobinopathies, Excluding Thalassemias: Basic and Translational: Poster II
Session Date: December 7, 2025
Session Time: 6:00 PM - 8:00 PM
Presentation Time: 6:00 PM - 8:00 PM
Room: OCCC – West Halls B3-B4
Publication Number: 2937
Title: **Cerebral and peripheral microvascular function in sickle cell disease: Blood flow and oxygen extraction via diffuse correlation and Near-Infrared spectroscopy**
Presenter: Dheemant Jallepalli